Weighted-Proj componentwise quotient prerequisite

Bounded Lean proof packet · 25 October 2026 · arbitrary field and coefficients

The pinned Lean file proves two prerequisites for a graded quotient of the weight (1,3,1) sextic. It does not construct the graded quotient, identify the degree-zero localization, or prove a projective chart.

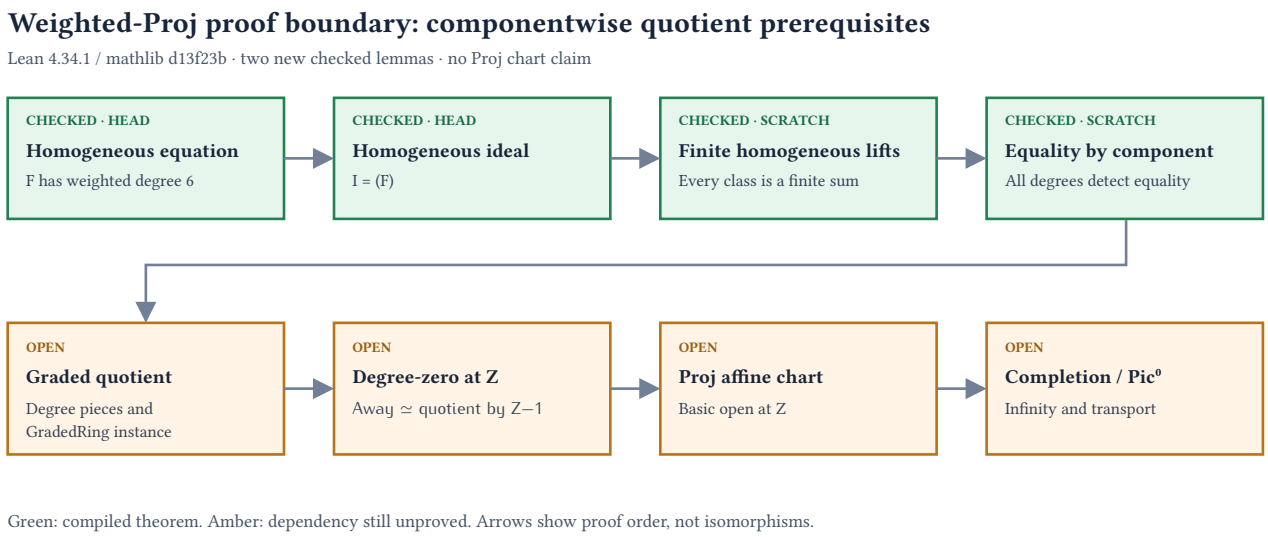

Figure 1. Green boxes are compiler-checked statements; amber boxes remain open proof obligations. Arrows indicate dependency order.

Checked statements

- **Finite homogeneous lifts:** every ordinary quotient class has a finite sum of images of weighted-homogeneous polynomial components.
- **Componentwise equality:** quotient representatives agree exactly when their weighted components agree in every degree.

Both theorems are in `WeightedProjDegreeZeroScratch.lean`. Lean 4.34.1 with mathlib `d13f23b` exited 0. The printed axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`.

Exact evidence

Source SHA-256: `2026d342f368bd89bc60f16267edbdd8d6299c5401d6c625c539202aa3384377`

Compiler log SHA-256: `d8ebceae0a02c6ce59ce3269cbb9cc7cb07ba1b8e4e9face104c67ad73414317`

The replay command, imported-module path, theorem statements, and checked/open rows are in `README.md` and `componentwise-status.csv`. No performance measurement is reported here.

Open boundary

The next steps are quotient degree pieces and a `GradedRing` instance, then an equivalence from the degree-zero homogeneous localization away from `Z` to the already established ordinary `Z = 1` quotient. Only after those steps can mathlib's `Proj.basicOpenIsoSpec` be specialized to this curve.